From "Truth Topology" to Readout Bottlenecks: Negative Replication on GSM8K and a Positive Representation–Decoder Dissociation in Procedural Micro-Worlds

Marcel
Independent Researcher

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Abstract

This paper reports a multi-stage study of hidden-state geometry for correctness prediction in language-model reasoning. The original hypothesis was ambitious: perhaps global topological summaries of hidden-state traces could provide a robust signal of correctness or "truth." That hypothesis did not survive stronger checks. On GSM8K-style reasoning traces, dynamic and windowed H_0 features were weak, often degenerate, and did not outperform simple controls. A separate fixed-decoding GSM8K branch yielded a stronger operational result: non-convergence, as measured by token-cap termination, was the dominant predictor of error.

The main positive result emerged only after changing the task. We introduced a procedural micro-world benchmark with nonce vocabularies, held-out templates, and exact tri-label semantics over `True`/`False`/`Unknown`. In that setting, a consistent dissociation appeared across Qwen and Gemma families: decoder behavior systematically under-expressed `Unknown`, yet verdict-region hidden-state probes recovered substantial `Unknown` signal on held-out worlds. This dissociation survived prompt-path controls, constrained label decoding, base-vs-instruct checks, verdict-step logit analysis, and layer sweeps. The paper's final claim is therefore narrower and stronger than the original repository framing: semantic non-entailment is internally represented, but often under-realized by decoder readout.

1 Scope and Final Claim

This project began as a search for a global internal marker of correctness. The final evidence supports a different claim:

> In a held-out procedural semantic task, non-entailment (`Unknown`) is represented in hidden states but systematically under-realized by decoder outputs.

The paper therefore has three parts:

1. a negative replication record for global-topology correctness signals on GSM8K-style traces,
2. a fixed-decoding convergence analysis showing cap-linked non-convergence as a dominant error mode, and

3. a positive result on a procedural micro-world benchmark showing a replicated representation–decoder dissociation for non-entailment.

The main contribution is *not* a universal scalar "topology of truth." The stronger result is a readout-bottleneck interpretation supported by multiple controls.

2 Why the Original Idea Failed

The original working hypothesis was that correct reasoning might have a cleaner global geometric or topological signature than incorrect reasoning. The basic recipe was:

1. run a small model on a reasoning task,
2. collect token-level hidden states,
3. treat those hidden states as a point cloud,
4. compute topological summaries such as connected-component fragmentation (H_0) and loop structure (H_1), and
5. ask whether those summaries predict correctness.

The hypothesis was attractive, but the experiments exposed three problems.

Length confounding. Prefix-based features changed largely because more points accumulated, not because the underlying reasoning became more truthful.

Degenerate window geometry. Fixed-size window preprocessing, especially per-window standardization and whitening, often collapsed windows into nearly geometry-equivalent objects. This made several windowed H_0 summaries almost pure functions of window size.

Task contamination. On natural-language math traces, hidden states reflect too many entangled factors at once: reasoning quality, verbosity, formatting drift, stop conditions, and non-convergence. This makes a single global topology scalar too blunt to isolate correctness.

The failure of the first branch is itself important. It prevented a misleading positive claim and forced a cleaner task design.

3 Experimental Trajectory

3.1 Phase A: Global Topology on GSM8K (Negative)

Small paired runs on Qwen3.5-0.8B and Qwen3.5-2B produced weak dynamic-H_0 signals on non-capped 2B traces. Representative pilot numbers were:

- `h0_entropy_final AUC = 0.55`

- `peak_h0_entropy AUC = 0.55`
- `auc_h0_entropy_per_token AUC = 0.50`
- `max_delta_h0 AUC = 0.60`

A corrected fixed-window test did worse:

- controls-only AUC: 0.60
- corrected-topology-only AUC: 0.20
- controls plus corrected-topology AUC: 0.20

These numbers are not compatible with a strong correctness-prediction claim.

3.2 Phase B: Fixed-Decoding GSM8K (Convergence Result)

A second branch held decoding fixed for Qwen3.5-2B and shifted focus from topology to operational failure mode. The key variables were:

$$C = \mathbf{1}[\text{run terminated by } \texttt{max_new_tokens}], \qquad Y = \mathbf{1}[\text{answer is wrong}].$$

On a within-question batch of $n = 180$ samples:

- capped: 87 samples (75 wrong / 12 correct)
- EOS: 93 samples (17 wrong / 76 correct)
- AUC($C \to Y$): 0.839
- odds ratio (wrong given cap): 26.40

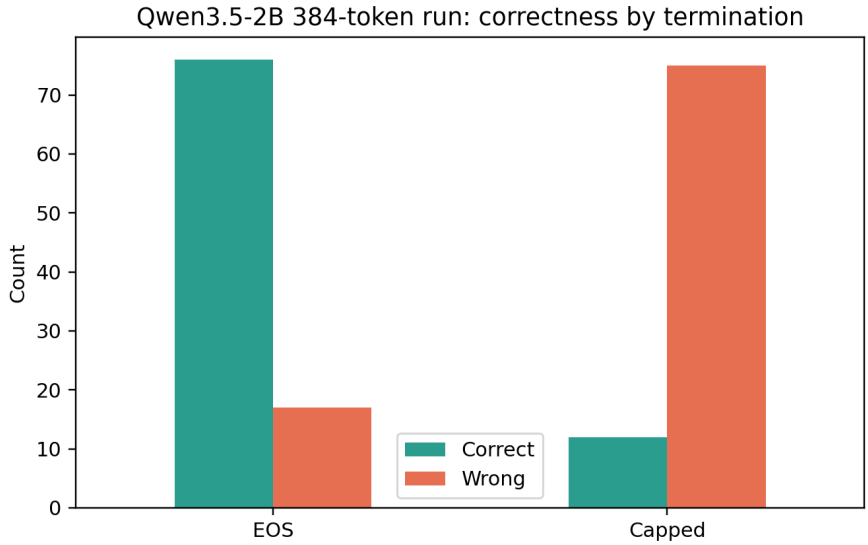

Figure 1: Fixed-config Qwen3.5-2B run: correctness split by termination mode.

The odds ratio calculation is:

$$\mathrm{OR} = \frac{\frac{75}{12}}{\frac{17}{76}} \approx 26.40.$$

A matched sensitivity rerun at 640 tokens for the originally capped cases showed that both truncation and deeper non-convergence mattered: some failures were rescued by a longer budget, but many remained wrong even after receiving more room to continue.

3.3 Phase C: Procedural Micro-World Semantics (Main Positive Result)

The project became scientifically cleaner only after replacing benchmark reasoning with a procedurally generated semantic task. Each generated world contains:

- a latent structured state,
- nonce entity names, attribute names, and relation names,
- held-out lexicons across train/dev/eval,
- held-out paraphrase templates across train/dev/eval,
- exact tri-label truth conditions over `True`, `False`, and `Unknown`.

Each world yields 72 examples: 9 propositions times 8 paraphrases. The main sweeps used 20 train worlds and 20 eval worlds.

4 Task Definition

Let a world be a finite structured state

$$w = (E, A, R),$$

where E is a set of entities, A denotes unary predicates (attributes), and R denotes binary predicates (relations). A proposition ϕ is rendered into natural language by a paraphrase template. Its semantic status is one of three classes:

$$y \in \{\texttt{True}, \texttt{False}, \texttt{Unknown}\}.$$

The labels are exact:

- `True`: the world entails ϕ,
- `False`: the world entails $\neg\phi$,
- `Unknown`: neither ϕ nor $\neg\phi$ is entailed.

This matters because `Unknown` is not a stylistic hesitation label. It is exact non-entailment under the generator.

4.1 Worked Micro-World Example

Table 1 shows one illustrative world/query slice.

Table 1: Worked micro-world example (illustrative format).

World fact set	Query statement	Gold label
`mep is falm; grel is not falm; nalo foshes sop`	`The object mep has property falm.`	True
`mep is falm; grel is not falm; nalo foshes sop`	`The object grel has property falm.`	False
`mep is falm; grel is not falm; nalo foshes sop`	`The relation fosh holds from mep to nalo.`	Unknown

5 What Is Stored for Each Example

For each example, the pipeline stores:

- the rendered prompt,
- the gold label,
- the emitted label,
- correctness and parse status,
- token IDs,
- selected hidden states,
- verdict-step logits for canonical label tokens.

The key state locations are:

- `final_prompt`
- `prompt_tail_mean`
- `verdict_token`
- `verdict_span_mean`

These are the local decision states of the model near label emission.

6 Methods

6.1 Linear Probe

Take a hidden state from one example at one chosen location:

$$h \in \mathbb{R}^d.$$

Each example provides a pair (h_i, y_i), where $y_i \in \{1, 2, 3\}$ corresponds to `True`, `False`, or `Unknown`.

A linear probe is

$$W \in \mathbb{R}^{3\times d}, \qquad b \in \mathbb{R}^3$$

with class scores

$$s = Wh + b$$

and prediction

$$\hat{y} = \arg\max_c s_c.$$

Training minimizes multiclass cross-entropy:

$$\mathcal{L}(W, b) = -\sum_i \log \frac{\exp((Wh_i + b)_{y_i})}{\sum_c \exp((Wh_i + b)_c)}.$$

This probe is intentionally weak. If it succeeds, the information is already arranged in hidden space in a directly readable linear form.

6.2 Within-World Geometry Gap

For one world and one state key, with distance $d(\cdot, \cdot)$:

$$D_{\text{same}} = \mathbb{E}[d(h_i, h_j) \mid y_i = y_j],$$

$$D_{\text{diff}} = \mathbb{E}[d(h_i, h_j) \mid y_i \neq y_j].$$

Define the world-level gap:

$$\Delta = D_{\text{diff}} - D_{\text{same}}.$$

A positive Δ means same-label states are more tightly organized than different-label states.

6.3 Verdict-Step Label-Logit Metrics

Let verdict-step logits for canonical label tokens be

$$\ell_T,\ \ell_F,\ \ell_U.$$

Then

$$p(c \mid h) = \frac{e^{\ell_c}}{e^{\ell_T} + e^{\ell_F} + e^{\ell_U}},$$

with the Unknown competitiveness margin

$$m_U = \ell_U - \max(\ell_T, \ell_F).$$

If $m_U < 0$, Unknown is under-ranked against the strongest non-Unknown candidate.

6.4 Layer Sweeps

For each layer ℓ, evaluate the same probe protocol and report

$$R_U^{(\ell)} = \text{Unknown recall of the probe at layer } \ell.$$

This reveals where non-entailment is maximally linearly recoverable in the network.

7 Main Micro-World Results

7.1 Decoder Behavior vs Hidden-State Recoverability

Table 2 summarizes the central result on held-out eval worlds.

Table 2: Decoder behavior vs verdict-token probe on held-out eval worlds ($n = 1440$ examples per model).

Model	Decoder accuracy	Decoder Unknown recall	Probe Unknown recall
Qwen3.5-2B	0.436	0.000	0.738
Qwen3.5-4B (no-think)	0.478	0.000	0.129
Gemma-3-4B-it (no-think)	0.397	0.013	0.563

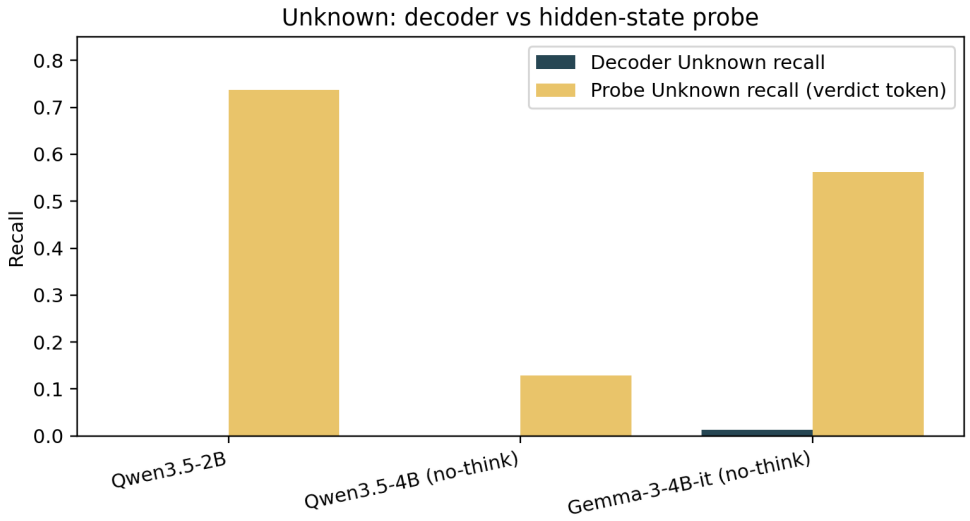

Figure 2: Unknown recall: decoder outputs vs verdict-token probe.

The key pattern is not raw accuracy. It is the gap between emitted-`Unknown` behavior and hidden-state recoverability of `Unknown`.

7.2 World-Level Geometry Consistency

World-level same-vs-different label distance gaps were positive in every evaluated world for main state keys in both Qwen3.5-2B and Gemma-3-4B-it.

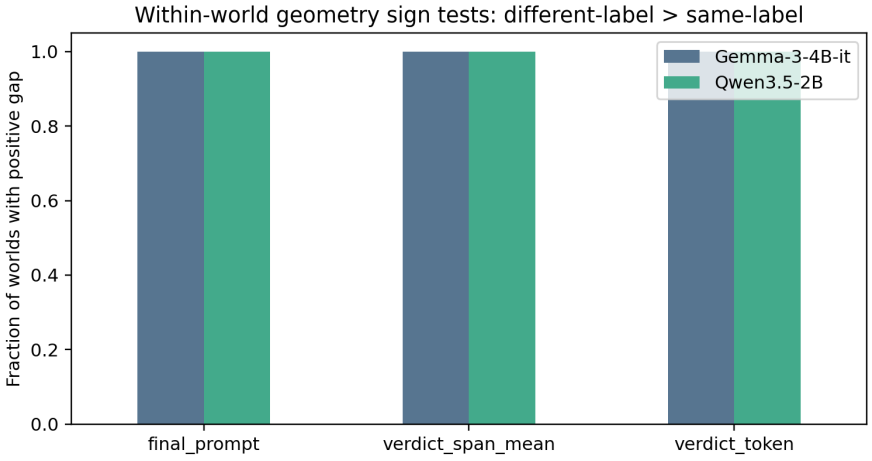

Figure 3: Fraction of worlds with positive same-vs-different label distance gap.

7.3 Cross-Family Replication

The dissociation replicates across Qwen and Gemma:

1. decoder outputs under-express `Unknown`,

2. hidden states still encode `Unknown`,
3. the gap survives held-out worlds, held-out lexicons, and held-out templates.

8 Mechanistic Controls

8.1 Constrained Decoding

To test whether free-form decoding alone caused the issue, decoding was constrained to {`True`, `False`, `Unknown`}. This did not repair Unknown collapse in the main models.

8.2 Prompt-Path Control

Raw-prompt controls were used to rule out chat-template-only explanations. The core dissociation remained.

8.3 Base vs Instruct

Gemma base initially showed severe parse failures under an unsuitable prompt path. After repairing prompt format with a base-specific label format, the parse confound disappeared. Yet base still showed decoder Unknown collapse while probes recovered substantial Unknown signal.

9 Verdict-Step Label-Logit Analysis

For gold-`Unknown` decoder failures:

- Gemma-3-4B-it: mean $P(\texttt{Unknown}) = 0.192$, mean margin $m_U = -1.293$,
- Gemma-3-4B-pt (basefmt): mean $P(\texttt{Unknown}) = 0.177$, mean margin $m_U = -1.045$.

So Unknown is often present but not competitive enough at final label-token competition.

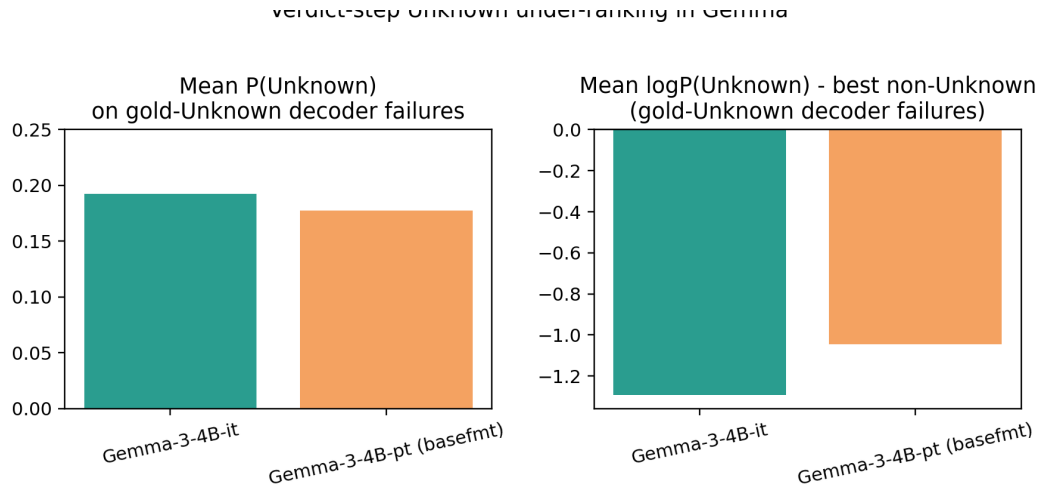

Figure 4: Unknown under-ranking at verdict step on gold-Unknown decoder failures.

10 Layer Sweeps

The best Unknown recall by state family was:

- Gemma-3-4B-it: `prompt_last` 0.779 at layer 1, `verdict_token` 0.742 at layer 10
- Gemma-3-4B-pt (basefmt): `prompt_last` 0.825 at layer 29, `verdict_token` 0.733 at layer 28

This shows strong recoverable Unknown signal exists internally in both instruct and base variants, even when emitted behavior still collapses that class.

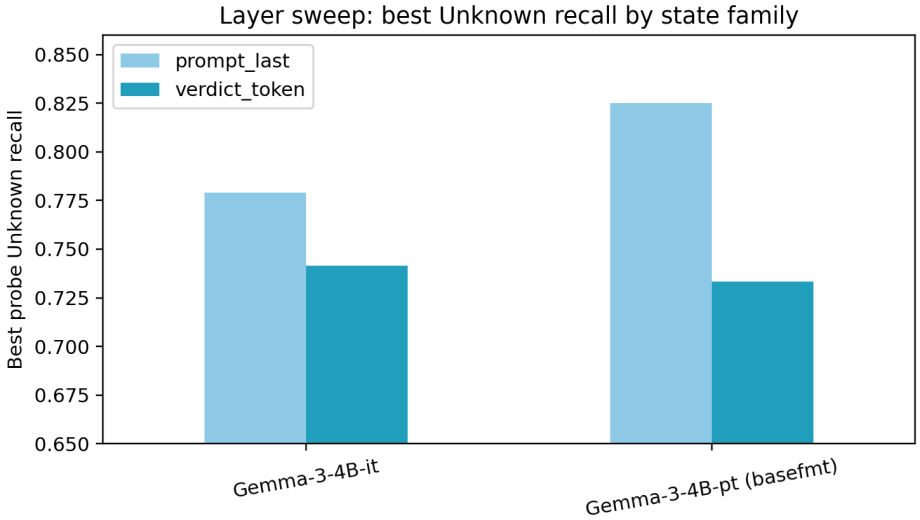

Figure 5: Best Unknown recall from layer sweeps by state family.

11 Interpretation

The combined evidence supports a readout-bottleneck interpretation:

1. semantic non-entailment is encoded internally,
2. decoder outputs under-express that class,
3. the mismatch is replicated across model families and controls,
4. verdict-step logits under-rank Unknown,
5. layer sweeps show strong internal signal upstream of failed readout.

This is stronger and narrower than the original repository framing. Results do *not* support a universal "truth topology" scalar.

12 Why This Should Survive Review

The result survives the obvious reviewer attacks:

- **Memorization objection**: nonce vocabularies, held-out lexicons, held-out templates.
- **Single-family objection**: replicated in both Qwen and Gemma.
- **Template objection**: raw-prompt controls.
- **Parse objection**: Gemma base rerun with repaired prompt format.
- **Free-decoding objection**: constrained decoding tested.
- **"No internal Unknown" objection**: probes, geometry, logits, and layer sweeps all counter it.

No single analysis carries the paper; strength comes from triangulation.

13 Practical Artifact Map

Key outputs used in the paper:

- convergence branch: `artifacts/qwen35_2b_within_question/analysis/`
- cap sensitivity: `artifacts/qwen35_2b_cap_sensitivity_640/analysis/`
- cross-model decoder comparisons: `artifacts/micro_world_v1/comparison_decoder_*.csv`
- probe comparisons: `artifacts/micro_world_v1/comparison_probe_states_*.csv`
- constrained-decoding comparisons: `artifacts/micro_world_v1/comparison_decoder_constrained_*.csv`
- verdict-step logits: `artifacts/micro_world_v1/label_logits_*/`
- layer sweeps: `artifacts/micro_world_v1/layer_sweep_*/`

14 Limitations

- The micro-world benchmark is synthetic, even though controlled and compositional.
- Model coverage is small-to-mid scale; larger-scale behavior is open.
- Probe recoverability does not prove causal use by decoder policy.
- The exact source of readout failure is not fully decomposed; it may involve calibration, label competition, or post-training behavior shaping.

15 Reproducibility

Locked dependency snapshot: `repro/requirements.lock.txt`

Figure generation:

```
python3 paper/scripts/make_figures.py
```

Compile:

```
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

16 Conclusion

The original "truth topology" objective was too broad for the observed evidence. The final validated contribution is a replicated representation–decoder dissociation: semantic non-entailment is strongly encoded in hidden states but systematically under-surfaced at output time. The correct framing is no longer a search for a single topology-of-truth scalar, but a study of how semantic uncertainty is preserved internally and lost at decoder readout.

A Appendix A: Full Reproduction Commands

```
# Phase A dynamic topology
python3 scripts/analyze_paired_dynamic.py \
  --artifact-08b artifacts/paired_qwen35_08b_10 \
  --artifact-2b artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/paired_dynamic_qwen35

# Phase A corrected window test
python3 scripts/analyze_2b_window_corrected.py \
  --artifact-dir artifacts/paired_qwen35_2b_10 \
```

```
  --out-dir artifacts/qwen35_2b_window_corrected

# Phase B fixed-decoding run
python3 scripts/run_multi_sample_questions.py \
  --model-id Qwen/Qwen3.5-2B \
  --question-start 0 --question-count 20 --samples-per-question 9 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 384 \
  --artifact-dir artifacts/qwen35_2b_within_question

# Phase B cap sensitivity
python3 scripts/run_cap_sensitivity.py \
  --model-id Qwen/Qwen3.5-2B \
  --source-artifact-dir artifacts/qwen35_2b_within_question \
  --out-dir artifacts/qwen35_2b_cap_sensitivity_640 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 640

# Phase C dataset
python3 scripts/generate_micro_world_dataset.py \
  --out-dir artifacts/micro_world_v1/dataset \
  --seed 1729 --train-worlds 100 --dev-worlds 25 --eval-worlds 100 \
  --props-per-world 9 --paraphrases-per-prop 8
```